

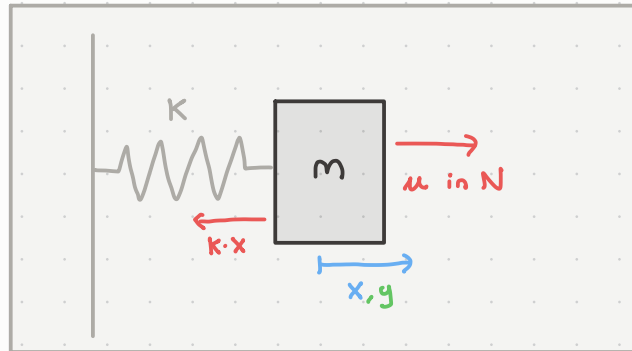
Exercise 3



Goal Today: Linearization, System Classification

Last week's cliffhanger: spring model vs drone model

- ① Determine inputs, outputs



- ② Find physical equation (Bathtub law)

$$\ddot{x} = u/m - k/m \cdot x_0$$



- ③ Write in standard form

$$\begin{cases} \dot{x} = F(x, u) \\ y = g(x, u) \end{cases}$$

$$\begin{pmatrix} \dot{x}_0 \\ \dot{x}_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ u/m - k/m \cdot x_0 \end{pmatrix}$$

$$y = x_0$$



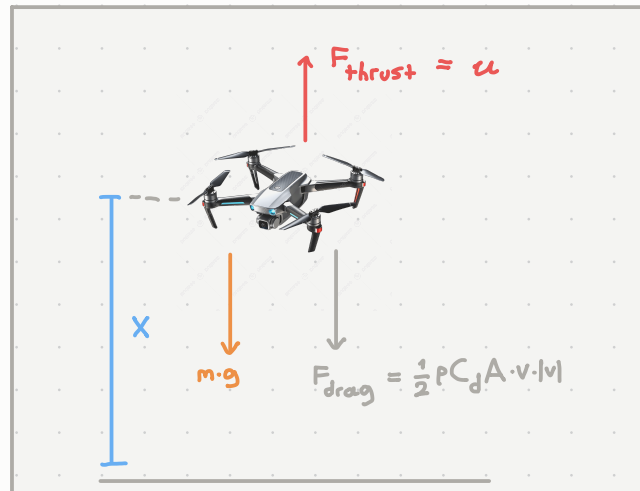
- * Write in State-Space Form

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

$$\begin{pmatrix} \dot{x}_0 \\ \dot{x}_1 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -k/m & 0 \end{pmatrix}}_A \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1/m \end{pmatrix}}_B \cdot u$$

$$y = \underbrace{(1 \quad 0)}_C \begin{pmatrix} x_0 \\ x_1 \end{pmatrix}$$

- ① Determine inputs, outputs



- ② Find physical equation (Bathtub law)

$$\ddot{x} = u/m - \frac{1}{2} \rho C_d A \cdot \dot{x}_1 |\dot{x}_1| - g$$



- ③ Write in standard Form

$$\begin{pmatrix} \dot{x} = F(x, u) \\ y = g(x, u) \end{pmatrix}$$

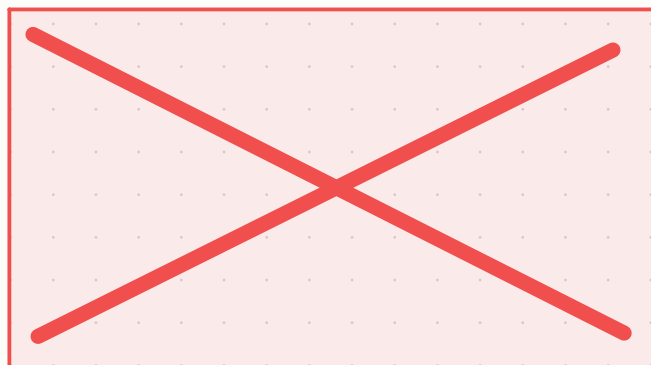
$$\begin{pmatrix} \dot{x}_0 \\ \dot{x}_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ u/m - \frac{1}{2} \rho C_d A \cdot \dot{x}_1 |\dot{x}_1| - g \end{pmatrix}$$

$$y = x_0$$



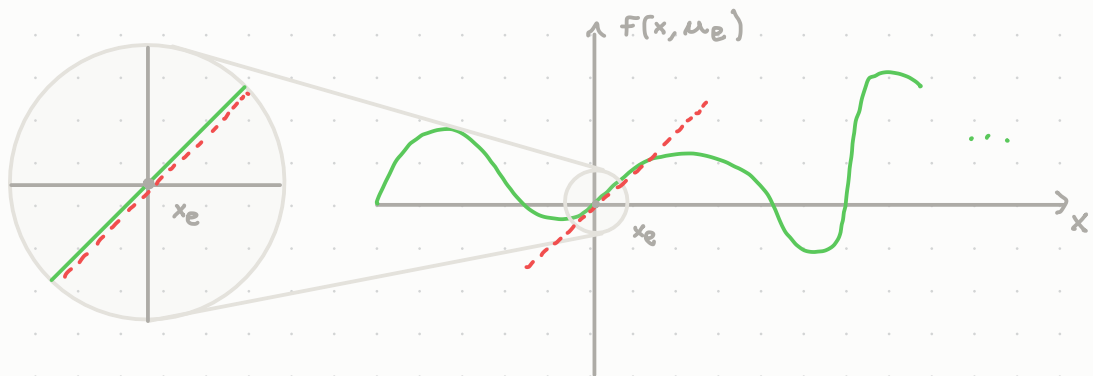
- * Write in State-Space Form

$$\begin{pmatrix} \dot{x} = Ax + Bu \\ y = Cx + Du \end{pmatrix}$$

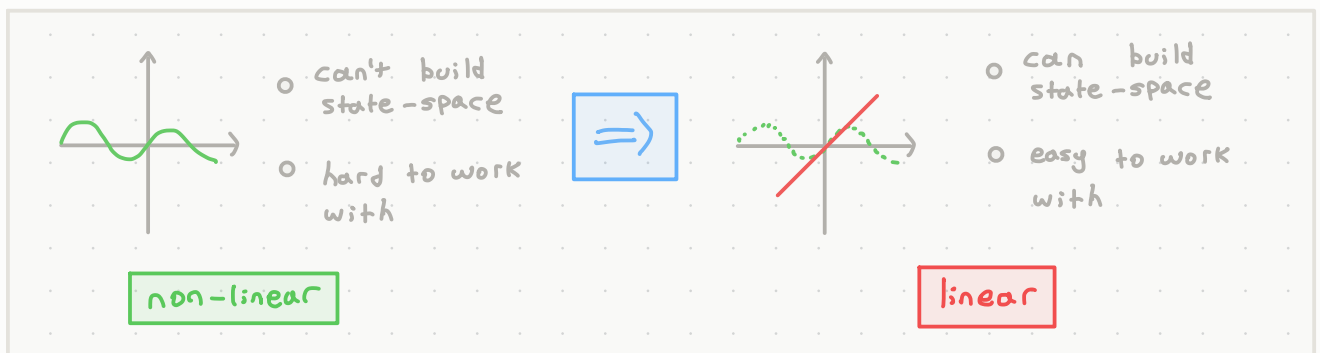


⚠ We need a linear model to write it in State-Space form. The drone's dynamics are non-linear, but we can linearize it near a point we care about

- Any smooth, non-linear curve looks like a linear line if we zoom in far enough at a point:



- In a tight neighborhood around this point the tangent approximates our function very well. This will be the idea of linearization:



⚠ Although when we leave close proximity of this point, the linearized system becomes inaccurate

- So two questions arise:

- ① Which point do we choose to approximate around?
- ② How do we build the tangent around this point?

? Your turn: equilibrium point of our drone

- Given the dynamics:

$$\begin{pmatrix} \dot{x}_0 \\ \dot{x}_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ \mu/m - \frac{1}{2m} \rho C_d A \cdot x_1 |x_1| - g \end{pmatrix}$$
$$y = x_0$$

Find $x_e = \begin{pmatrix} x_{0,e} \\ x_{1,e} \end{pmatrix}$ and μ_e

- Think why the x_e we found makes sense for a drone

② How do we build the **tangent** around this point?

Idea: use the **Taylor-Approximation** of the function

$$F(x, \mu) = F(x_e, \mu_e) + \left. \frac{\partial F}{\partial x} \right|_{x_e, \mu_e} \cdot (x - x_e) + \left. \frac{\partial F}{\partial \mu} \right|_{x_e, \mu_e} \cdot (\mu - \mu_e) + \dots$$



$$F(x, \mu) = F(x_e, \mu_e) + \left. \frac{\partial F}{\partial x} \right|_{x_e, \mu_e} \cdot \underbrace{(x - x_e)}_{\delta x} + \left. \frac{\partial F}{\partial \mu} \right|_{x_e, \mu_e} \cdot \underbrace{(\mu - \mu_e)}_{\delta \mu} + \dots$$

we picked an equilibrium point...

Higher order terms don't interest us. We want a linear model.



$$F(x, \mu) = \left. \frac{\partial F}{\partial x} \right|_{x_e, \mu_e} \cdot \delta x + \left. \frac{\partial F}{\partial \mu} \right|_{x_e, \mu_e} \cdot \delta \mu$$



we usually drop the δ 's

$$F(x, \mu) = \left. \frac{\partial F}{\partial x} \right|_{x_e, \mu_e} \cdot x + \left. \frac{\partial F}{\partial \mu} \right|_{x_e, \mu_e} \cdot \mu$$

- Remember our standard form was:

$$\begin{aligned}\dot{x} &= f(x, u) \\ y &= g(x, u)\end{aligned}$$

- So if $g(x, u)$ is also non-linear, we linearize it in the same way:

$$g(x, u) = \left. \frac{\partial g}{\partial x} \right|_{x_e, u_e} \cdot x + \left. \frac{\partial g}{\partial u} \right|_{x_e, u_e} \cdot u$$

- And there we have it! Our linearized model in state-space form!

$$\begin{aligned}\dot{x} &= \overset{A}{\left. \frac{\partial f}{\partial x} \right|_{x_e, u_e}} \cdot x + \overset{B}{\left. \frac{\partial f}{\partial u} \right|_{x_e, u_e}} \cdot u \\ y &= \overset{C}{\left. \frac{\partial g}{\partial x} \right|_{x_e, u_e}} \cdot x + \overset{D}{\left. \frac{\partial g}{\partial u} \right|_{x_e, u_e}} \cdot u\end{aligned}$$

where:

$$A = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

$$B = \begin{pmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \\ \vdots \end{pmatrix}$$

and

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} f_1(x, u) \\ f_2(x, u) \\ \vdots \end{pmatrix}$$

$$C = \left(\frac{\partial g}{\partial x_1} \quad \frac{\partial g}{\partial x_2} \quad \dots \right)$$

$$D = \frac{\partial g}{\partial u}$$

$$y = g(x, u)$$

Example: Our drone

$$\begin{pmatrix} \dot{x}_0 \\ \dot{x}_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ \mu/m - \frac{1}{2m} \rho C_d A \cdot x_1 |x_1| - g \end{pmatrix}$$
$$y = x_0$$

○ Linearize the system around $\begin{pmatrix} x_e = (0, h) \\ \mu_e = mg \end{pmatrix}$.

$$A = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \\ \vdots \end{pmatrix}$$

$$B = \begin{pmatrix} \frac{\partial f_1}{\partial \mu} \\ \frac{\partial f_2}{\partial \mu} \\ \vdots \end{pmatrix}$$

$$C = \begin{pmatrix} \frac{\partial g}{\partial x_1} & \frac{\partial g}{\partial x_2} & \dots \end{pmatrix}$$

$$D = \frac{\partial g}{\partial \mu}$$

? Your turn:

Problem: You are given the following nonlinear system with state vector $x(t) \in \mathbb{R}^2$, input $u(t) \in \mathbb{R}$ and output $y(t) \in \mathbb{R}$:

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} x_1(t) + \sin(x_2(t) + u(t)) \\ x_1^2(t) + e^{u(t)} \end{bmatrix},$$

$$y(t) = x_1(t) - \cos(u(t)).$$

Linearize the system around the equilibrium point (x_e, u_e) where

$$x_e = \begin{bmatrix} x_{1,e} \\ x_{2,e} \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix},$$

$$u_e = 0.$$

Derive the (A, B, C, D) matrices of the state-space representation of the linearization of the nonlinear system around (x_e, u_e) .

Q6 (1 Points) What is the A matrix of the linearization of the above system around the equilibrium (x_e, u_e) ?

$$A =$$

Q7 (0.5 Points) What is the B matrix of the linearization of the above system around the equilibrium (x_e, u_e) ?

$$B =$$

Q8 (0.5 Points) What are the C and D matrices of the linearization of the above system around the equilibrium (x_e, u_e) ?

$$C =$$

$$D =$$

System Classification

- In Control systems, we classify systems ($y = \sum u$) with 4 categories (questions).

① Linearity — "Do sums and scalings go through?"

- For a system to be linear two conditions have to be fulfilled:

→ Additivity: $\sum (u_1 + u_2) = \sum u_1 + \sum u_2$

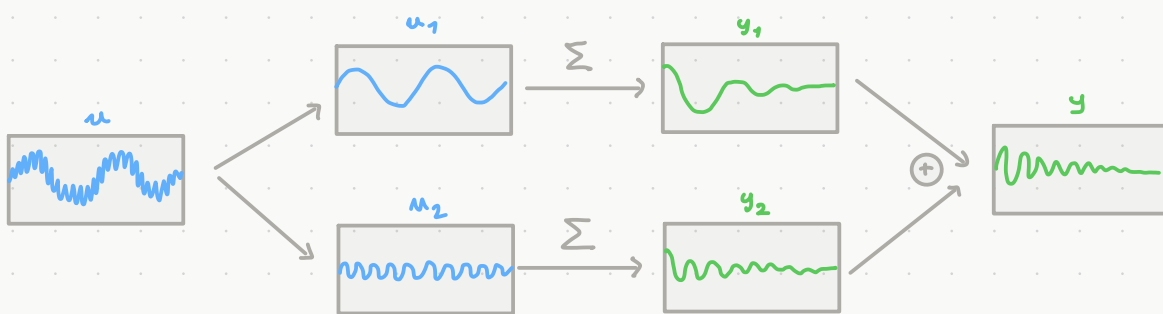
→ Homogeneity: $\sum Ku = K \sum u$, $K \in \mathbb{R}$

we can summarize both to:

$$\sum (\alpha u_1 + \beta u_2) = \alpha \sum u_1 + \beta \sum u_2 = \alpha \cdot y_1 + \beta \cdot y_2$$

- Example: $y(t) = t^2 \cdot u(t-1)$

- This property is incredibly useful i.r.l. We can decompose a complicated input into easier signals and compute the output for each:



$$\sum u = \sum (u_1 + u_2) = y_1 + y_2 = y$$

② Your Turn: $y_1 = \frac{d}{dt} u(t)$ $y_2 = u^2(t)$ $y_3 = \int_{-\infty}^t u(\tau) d\tau$

② Causality — "Does the output need the future?"

- A system is causal only if future inputs do not affect the present output. All realizable machines are causal.

③ Your turn:

Causal

Non-Causal

$$y(t) = u(t-\alpha), \alpha < 0$$

$$y(t) = u(t-\tau), \tau > 0$$

$$y(t) = \cos(3t+1) \cdot u(t-1)$$

$$y(t) = \int_{-\infty}^{\infty} u(\tau) d\tau$$

③ Static (memoryless) — "Is the present $u(t)$ enough?"

- A system is static if $y(t)$ only depends on the present $u(t)$ (no past nor future u).
- Systems described by ODE's are dynamic:

$$\begin{aligned} \dot{x} &= u(t) \\ y &= x \end{aligned} \Rightarrow y = \int_{-\infty}^t u(\tau) d\tau$$

③ Your turn:

Static

Dynamic

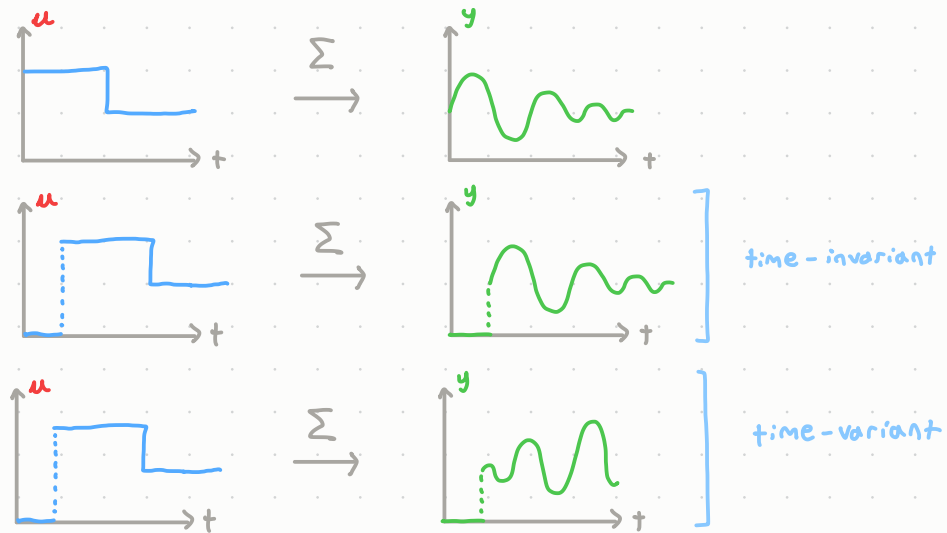
$$y(t) = 2^{-(t+1)} \cdot u(t)$$

$$y(t) = \dot{u}(t)$$

$$y(t) = u(t^2)$$

④ Time-invariant? — "Does it matter when I do it?"

- A time invariant system will always have the same output to a certain input, no matter when the input is applied:



- It is tested by:

$$\begin{aligned}\tilde{u}(t) &= u(t-\tau) \\ \tilde{y}(t) &\stackrel{?}{=} y(t-\tau)\end{aligned}$$

If yes,
time-invariant

- Example: $y(t) = u(3t+2)$

⑤ Your Turn:

$$\begin{aligned}y(t) &= t \cdot u(t) \\ y(t) &= u(t^2) \\ y(t) &= \int_{-\infty}^t u(\tau) d\tau\end{aligned}$$

Time-inv. Time-var.

Order of a system:

- The order of a system is the dimension of the state vector x .